Lesson Planning

Complete Lesson Plan Book

Class: Class 9

Subject: Computer Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Fundamentals of Computer

Learning Objectives

• Define a computer and describe its characteristics.

• Explain the IPO cycle with examples.

• Classify computers by type and size.

• Describe the role of hardware and software.

Key Concepts

• Computer: electronic device that processes data

• Characteristics: speed, accuracy, diligence, versatility

• IPO cycle: Input, Process, Output

• Types: digital, analog, hybrid; super, mainframe, mini, micro

• Hardware vs software

Teaching Methodology (Step by Step)

1. Ask how a calculator 'knows' answers: introduce data processing.
2. Draw the IPO cycle with a mobile-phone example.
3. Show pictures of computer types from supercomputer to smartphone.
4. Let students list hardware vs software examples in two columns.

Classroom Activities

• Classify ten given devices into computer types.

• Draw the IPO cycle for three daily-life machines.

Assessment Questions

1. Define a computer.
2. What is the IPO cycle?
3. Name four characteristics of a computer.
4. Give two examples each of hardware and software.

Homework

• Draw the IPO cycle with labelled examples.

• Write a paragraph on the uses of computers in education.

Time Allocation

4 periods of 40 minutes each

Unit 2: Number Systems

Learning Objectives

• Explain binary, octal, decimal and hexadecimal systems.

• Convert numbers between decimal and binary.

• Convert between binary, octal and hexadecimal.

• Perform binary addition and subtraction.

Key Concepts

• Base of a number system; place value

• Binary (base 2), octal (base 8), hexadecimal (base 16)

• Decimal to binary: repeated division

• Binary to decimal: positional weights

• Binary addition and subtraction rules

Teaching Methodology (Step by Step)

1. Use the 'doubling' game: 1, 2, 4, 8 fingers to introduce binary.
2. Demonstrate decimal-to-binary conversion step by step.
3. Practise binary addition with carry rules on the board.
4. Give timed conversion drills for speed.

Classroom Activities

• Convert your birth year to binary and hexadecimal.

• Binary addition race in teams on the board.

Assessment Questions

1. Convert 25 (decimal) to binary.
2. Convert 1011 (binary) to decimal.
3. Add 1011 + 0110 in binary.
4. What is the base of the hexadecimal system?

Homework

• Complete the conversion exercise (10 questions).

• Make a base-conversion reference card for the notebook.

Time Allocation

6 periods of 40 minutes each

DM

Unit 3: Boolean Algebra

Learning Objectives

• Define Boolean algebra and its two values.
• State the basic Boolean operators AND, OR, NOT.
• Apply the laws of Boolean algebra.
• Simplify simple Boolean expressions.

Key Concepts

• Boolean values: true/false, 1/0
• AND (.), OR (+), NOT (bar) operators
• Truth tables for basic operators
• Laws: identity, complement, commutative, distributive
• De Morgan's theorems

Teaching Methodology (Step by Step)

1. Use light-switch examples: both switches ON for AND.
2. Build truth tables together for AND, OR, NOT.
3. Prove one Boolean law with a truth table on the board.
4. Simplify expressions step by step with student input.

Classroom Activities

• Fill truth tables for given expressions in pairs.
• Simplify five Boolean expressions using the laws.

Assessment Questions

1. Write the truth table of the AND operator.
2. State De Morgan's first theorem.
3. Simplify A + A'B using Boolean laws.
4. Name the three basic Boolean operators.

Homework

• Complete the Boolean simplification exercise.
• Write all Boolean laws with one example each.

Time Allocation

5 periods of 40 minutes each

Unit 4: Logic Gates and Circuits

Learning Objectives

• Draw symbols and truth tables of basic logic gates.

• Explain universal gates NAND and NOR.

• Draw simple combinational circuits.

• Explain the half-adder circuit.

Key Concepts

• AND, OR, NOT gates: symbols and truth tables

• NAND and NOR as universal gates

• XOR and XNOR gates

• Combinational circuits from Boolean expressions

• Half adder: sum and carry

Teaching Methodology (Step by Step)

1. Show gate symbols as 'decision boxes' with input-output tables.
2. Build truth tables for NAND and NOR together.
3. Draw a circuit for a given expression step by step.
4. Explain the half adder using binary addition examples.

Classroom Activities

• Draw circuits for three given Boolean expressions.

• Simulate gates with switches or an online simulator if available.

Assessment Questions

1. Draw the symbol and truth table of the NOR gate.
2. Why are NAND and NOR called universal gates?
3. Draw the half-adder circuit.
4. Write the Boolean expression for an AND-OR circuit.

Homework

• Draw all gate symbols with truth tables in the notebook.

• Draw circuits for five given expressions.

Time Allocation

5 periods of 40 minutes each

Unit 5: Computer Memory

Learning Objectives

• Distinguish between primary and secondary memory.

• Explain RAM and ROM and their types.

• Describe common secondary storage devices.

• Compare memory units from bit to terabyte.

Key Concepts

• Memory hierarchy concept

• RAM: SRAM vs DRAM; volatile nature

• ROM: PROM, EPROM, EEPROM

• Secondary: HDD, SSD, USB, optical discs

• Units: bit, byte, KB, MB, GB, TB

Teaching Methodology (Step by Step)

1. Ask where a phone 'remembers' photos: introduce storage types.
2. Compare RAM and ROM in a table with student suggestions.
3. Show real storage devices if available; discuss capacities.
4. Practise converting between memory units.

Classroom Activities

• Rank ten storage devices by capacity and speed.

• Convert: 2 GB to MB, 512 MB to KB.

Assessment Questions

1. What is the difference between RAM and ROM?
2. Name three secondary storage devices.
3. How many bytes are in 1 KB?
4. Why is RAM called volatile memory?

Homework

• Draw the memory hierarchy pyramid with labels.

• Write uses of any four storage devices.

Time Allocation

4 periods of 40 minutes each

Unit 6: Data Communication

Learning Objectives

• Define data communication and its basic model.

• Explain transmission modes: simplex, half and full duplex.

• Describe transmission media: guided and unguided.

• Compare common network topologies.

Key Concepts

• Communication model: sender, message, medium, receiver, protocol

• Simplex, half-duplex, full-duplex with examples

• Guided media: twisted pair, coaxial, fibre optic

• Unguided media: radio, microwave, satellite

• Topologies: bus, star, ring, mesh

Teaching Methodology (Step by Step)

1. Act out sender-medium-receiver with a whispered message chain.
2. Use walkie-talkie vs phone-call examples for duplex modes.
3. Draw each topology on the board; discuss one failure point.
4. Compare fibre optic with copper cable in a table.

Classroom Activities

• Draw and label the four network topologies.

• Classify ten communication examples by transmission mode.

Assessment Questions

1. Draw the basic data communication model.
2. Give one example each of simplex and full-duplex communication.
3. Name two guided and two unguided media.
4. Which topology is most reliable? Why?

Homework

• Draw topologies with one advantage and one disadvantage each.

• Write a paragraph on the uses of fibre optic cable.

Time Allocation

5 periods of 40 minutes each

Unit 7: Computer Networks

Learning Objectives

• Define a network and explain its advantages.

• Classify networks as LAN, MAN and WAN.

• Explain the role of common network devices.

• Describe the uses of the Internet.

Key Concepts

• Network: connected computers sharing resources

• LAN, MAN, WAN: range and examples

• Devices: hub, switch, router, modem

• Internet services: email, web, e-learning

• Intranet vs extranet

Teaching Methodology (Step by Step)

1. Ask how the school's computers share one printer: introduce networks.
2. Compare LAN, MAN, WAN using city-scale examples.
3. Explain each device's job with a simple analogy.
4. Discuss safe and useful Internet habits.

Classroom Activities

• Design a LAN layout for a computer lab on paper.

• List ten Internet uses and mark educational ones.

Assessment Questions

1. Define a computer network.
2. Differentiate between LAN and WAN.
3. What is the function of a router?
4. Name any four Internet services.

Homework

• Draw a labelled LAN diagram for a school lab.

• Write five advantages of computer networks.

Time Allocation

4 periods of 40 minutes each

Unit 8: Computer Security and Ethics

Learning Objectives

• Define computer security and its goals.

• Describe malware: virus, worm, trojan horse.

• Explain measures to protect data: passwords, antivirus, backup.

• Discuss ethical issues: privacy, piracy, cybercrime.

Key Concepts

• CIA triad: confidentiality, integrity, availability

• Malware types and how they spread

• Protection: strong passwords, antivirus, firewall, backup

• Software piracy and copyright

• Cyber ethics: netiquette, privacy, responsible sharing

Teaching Methodology (Step by Step)

1. Share a real story of a hacked account to create awareness.
2. Classify malware examples into virus, worm, trojan.
3. Demonstrate what makes a password strong.
4. Debate: is sharing paid software with friends acceptable?

Classroom Activities

• Create a strong-password checklist and test sample passwords.

• Make a 'safe computing' poster for the classroom.

Assessment Questions

1. Name three types of malware.
2. What is the difference between a virus and a worm?
3. Give three rules of a strong password.
4. What is software piracy?

Homework

• Write ten safe-computing rules in the notebook.

• Write a paragraph on cyber ethics.

Time Allocation

4 periods of 40 minutes each